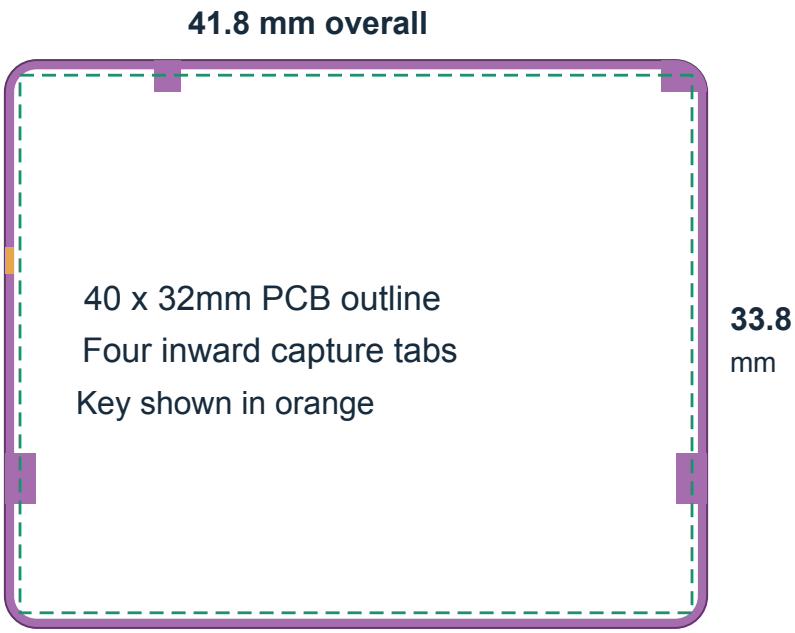
CT-RET-001 | Captive PCB frame / r10

PROTOTYPE — manufacturing, thermal fit, retention strength and ingress remain unqualified

Retainer plan



Frame thickness: 1.2mm; aperture: 40.7 x 32.7mm.

Board and its all-green Gerber remain unchanged.

Rigid enclosure: 57.4 x 49.4 x 21.0mm, plus docking key.

Assembly datums and checks

PCB support	Z=10.40mm; thickness 0.71–0.91mm
Guide / frame bottom	Z=11.50mm nominal
Lid stop plane	Z=12.85mm nominal
Maximum PCB top	Z=11.75mm in this model
SIM bridge	Z=10.70mm; seated gap ≥0.20mm
Component sweep	85 envelopes; minimum gap 0.205mm
Cable endpoint poses	27 cases; minimum body gap 0.299mm
Tab overlap	27 offset/yaw cases; ≥0.321mm² per tab
Manufacturing limits	Assigned requirements; inspect actual parts
Free play	Captive frame; zero movement is not claimed
Material	Insulating POM-C candidate; not approved

Factory closure and service

Insert SIM with power off and the board seated in the lifted carrier.

Fit the key, inspect all tabs and cables, then close without force.

Dimensions describe the ambient prototype. Verify manufacturing limits and thermal expansion before ordering.

The PCB remains captive within assigned bounds; shock, vibration, stiffness, wear and anti-rattle behavior need testing.

Cable checks sample endpoint movement. They do not prove every continuous deformation or physical flex life.

The pack is still an unapproved supplier request. Pack retention, harness/dock and firmware remain unfinished.

No supplier message, order or physical water/dust qualification is implied by this drawing.